Dengue viruses and promising envelope protein domain III-based vaccines.

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Dengue viruses are emerging mosquito-borne pathogens belonging to Flaviviridae family which are transmitted to humans via the bites of infected mosquitoes Aedes aegypti and Aedes albopictus. Because of the wide distribution of these mosquito vectors, more than 2.5 billion people are approximately at risk of dengue infection. Dengue viruses cause dengue fever and severe life-threatening illnesses as well as dengue hemorrhagic fever and dengue shock syndrome. All four serotypes of dengue virus can cause dengue diseases, but the manifestations are nearly different depending on type of the virus in consequent infections. Infection by any serotype creates life-long immunity against the corresponding serotype and temporary immunity to the others. This transient immunity declines after a while (6 months to 2 years) and is not protective against other serotypes, even may enhance the severity of a secondary heterotypic infection with a different serotype through a phenomenon known as antibody-depended enhancement (ADE). Although, it can be one of the possible explanations for more severe dengue diseases in individuals infected with a different serotype after primary infection. The envelope protein (E protein) of dengue virus is responsible for a wide range of biological activities, including binding to host cell receptors and fusion to and entry into host cells. The E protein, and especially its domain III (EDIII), stimulates host immunity responses by inducing protective and neutralizing antibodies. Therefore, the dengue E protein is an important antigen for vaccine development and diagnostic purposes. Here, we have provided a comprehensive review of dengue disease, vaccine design challenges, and various approaches in dengue vaccine development with emphasizing on newly developed envelope domain III-based dengue vaccine candidates.

Recent advances in the identification of the host factors involved in dengue virus replication.

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Dengue virus (DENV) belongs to the genus Flavivirus of the family Flaviviridae and it is primarily transmitted via Aedes aegypti and Aedes albopictus mosquitoes. The life cycle of DENV includes attachment, endocytosis, protein translation, RNA synthesis, assembly, egress, and maturation. Recent researches have indicated that a variety of host factors, including cellular proteins and microRNAs, positively or negatively regulate the DENV replication process. This review summarizes the latest findings (from 2014 to 2016) in the identification of the host factors involved in the DENV life cycle and Dengue infection.

Dengue virus infection in renal allograft recipients: a case series during 2010 outbreak.

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Dengue virus infection is an emerging global threat caused by Arbovirus, a virus from Flaviridiae family, which is transmitted by mosquitoes, Aedes aegypti and Aedes albopictus. Renal transplant recipients who live in the endemic zones of dengue infection or who travel to an endemic zone could be at risk of this infection. Despite multiple epidemics and a high case fatality rate in the Southeast Asian region, only a few cases of dengue infection in renal transplant recipients have been reported. Here, we report a case series of 8 dengue viral infection in renal transplant recipients. Of the 8 patients, 3 developed dengue hemorrhagic shock syndrome and died.

Dengue/DHF: an emerging disease in India.

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Dengue/DHF is an emergent disease in India and some parts of country are endemic and periodically contributing annual outbreaks of dengue/DHF. Dengue infection manifests as undifferentiated fever, dengue haemorrhagic fever (DHF) which leads to hospitalization large number of people in a localized area. There is high mortality and morbidity associated with the onset of each dengue outbreak leading to great socio-economic impact. The prevention and control of dengue outbreak depends upon the proper monitoring of the disease case through disease surveillance so as to ensure timely management of cases. Vector surveillance helps in the proper and timely implementation of emergency control measures against dengue vector i.e. Aedes aegypti. There is an urgent need for an effective diagnostic strategy for early diagnosis to shorten the illness duration, hospitalization time and the associated complications.

Dengue fever: a resurgent risk for the international traveler.

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The incidence of dengue fever, an acute febrile illness transmitted by the Aedes aegypti mosquito, is on the rise. High fever, severe headache, skin rash and a variety of constitutional symptoms are hallmarks of classic dengue fever. Dengue hemorrhagic fever, a severe manifestation associated with secondary infection, most often occurs in children. Treatment of classic dengue fever is supportive, whereas urgent rehydration therapy is often required in more severe forms. Community-based and personal strategies for avoiding the mosquito vector represent the best methods of prevention, although vaccine development programs are under way.

Disease prediction with multi-omics and biomarkers empowers case-control genetic discoveries in the UK Biobank.

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The emergence of biobank-level datasets offers new opportunities to discover novel biomarkers and develop predictive algorithms for human disease. Here, we present an ensemble machine-learning framework (machine learning with phenotype associations, MILTON) utilizing a range of biomarkers to predict 3,213 diseases in the UK Biobank. Leveraging the UK Biobank's longitudinal health record data, MILTON predicts incident disease cases undiagnosed at time of recruitment, largely outperforming available polygenic risk scores. We further demonstrate the utility of MILTON in augmenting genetic association analyses in a phenome-wide association study of 484,230 genome-sequenced samples, along with 46,327 samples with matched plasma proteomics data. This resulted in improved signals for 88 known ($P < 1 \times 10$-8) gene-disease relationships alongside 182 gene-disease relationships that did not achieve genome-wide significance in the nonaugmented baseline cohorts. We validated these discoveries in the FinnGen biobank alongside two orthogonal machine-learning methods built for gene-disease prioritization. All extracted gene-disease associations and incident disease predictive biomarkers are publicly available (http://milton.public.cgr.astrazeneca.com).

Loss of heterozygosity on chromosome 10 is more extensive in primary (de novo) than in secondary glioblastomas.

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Glioblastomas develop de novo (primary glioblastomas) or through progression from low-grade or anaplastic astrocytoma (secondary glioblastomas). There is increasing evidence that these glioblastoma subtypes develop through different genetic pathways. Primary glioblastomas are characterized by EGFR and MDM2 amplification/overexpression, PTEN mutations, and p16 deletions, whereas secondary glioblastomas frequently contain p53 mutations. Loss of heterozygosity (LOH) on chromosome 10 (LOH#10) is the most frequent genetic alteration in glioblastomas; the involvement of tumor suppressor genes, other than PTEN, has been suggested. We carried out deletion mappings on chromosome 10, using PCR-based microsatellite analysis. LOH#10 was detected at similar frequencies in primary (8/17; 47%) and secondary glioblastomas (7/13; 54%). The majority (88%) of primary glioblastomas with LOH#10 showed LOH at all informative markers, suggesting loss of the entire chromosome 10. In contrast, secondary glioblastomas with LOH#10 showed partial or complete loss of chromosome 10q but no loss of 10p. These results are in accordance with the view that LOH on 10q is a major factor in the evolution of glioblastoma multiform as the common phenotypic end point of both genetic pathways, whereas LOH on 10p is largely restricted to the primary (de novo) glioblastoma.

[Biology molecular of glioblastomas].

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Glioblastomas, the most frequent and malignant human brain tumors, may develop de novo (primary glioblastoma) or by progression from low-grade or anapalsic astrocytoma (secondary glioblastoma). The molecular alteration most frequent in these tumor-like types is the loss of heterozygosity on chromosome 10, in which several genes have been identified as tumors suppressor. The TP53/MDM2/P14arf and CDK4/RB1/ P16ink4 genetic pathways involved in cycle control are deregulated in the majority of gliomas as well as genes that promote the cellular division, EGFR. Finally the increase of growth and angiogenics factors is also involved in the development of glioblastomas. One of the objectives of molecular biology in tumors of glial ancestry is to try to find the genetic alterations that allow to approach better the classification of glioblastomas, its evolution prediction and treatment. The new pathmolecular classification of gliomas should improve the old one, especially being concerned about the oncogenesis and heterogeneity of these tumors. It is desirable that this classification had clinical applicability and integrates new molecular findings with some known histological features with pronostic value. In this paper we review the most frequent molecular mechanisms involved in the patogenesis of glioblastomas.

Loss of heterozygosity for 10q loci in human gliomas.

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Cytogenetic and RFLP studies have shown that chromosome 10 is frequently lost in tumor cells from glioblastomas, suggesting that a suppressor gene important in tumorigenesis is present on this chromosome. Forty-one tumors were examined for loss of heterozygosity at 23 loci on chromosome 10 to determine the smallest common deletion interval on this chromosome. Seven tumors did not lose heterozygosity for any of the markers. Twenty-three tumors lost an allele for all the informative loci. In 11 tumors heterozygosity was maintained at some loci and lost at other loci, indicating partial deletion of chromosome 10. The common region of deletion in these 11 tumors was located in 10q24-q26 between the markers pHUK-8 and pMCT122.2.

Loss of heterozygosity on chromosome 19 in secondary glioblastomas.

PMID: 10850866 | Indexed in PubMed

Glioblastomas develop rapidly de novo (primary glioblastomas) or slowly through progression from low-grade or anaplastic astrocytoma (secondary glioblastomas). Recent studies have shown that these glioblastoma subtypes develop through different genetic pathways. Primary glioblastomas are characterized by EGFR amplification/overexpression, PTEN mutation, homozygous p16 deletion, and loss of heterozygosity (LOH) on entire chromosome 10, whereas secondary glioblastomas frequently contain p53 mutations and show LOH on chromosome 10q. In this study, we analyzed LOH on chromosomes 19q, 1p, and 13q, using polymorphic microsatellite markers in 17 primary glioblastomas and in 13 secondary glioblastomas that progressed from low-grade astrocytomas. LOH on chromosome 19q was frequently found in secondary glioblastomas (7 of 13, 54%) but rarely detected in primary glioblastomas (1 of 17, 6%, $p = 0.0094$). The common deletion was 19q13.3 (between D19S219 and D19S902). These results suggest that tumor suppressor gene(s) located on chromosome 19q are frequently involved in the progression from low-grade astrocytoma to secondary glioblastoma, but do not play a major role in the evolution of primary glioblastomas. LOH on chromosome 1p was detected in 12% of primary and 15% of secondary glioblastomas. LOH on 13q was detected in 12% of primary and in 38% of secondary glioblastomas and typically included the RB locus. Except for 1 case, LOH 13q and 19q were mutually exclusive.